

other nodes to leave or move off the network faster. These attacks can be detected with resource testing, based on the assumption that vehicles have limited resources. Sybil attacks can be addressed using public key cryptography, where public keys are used to authenticate vehicles.

A node imitation attack: In this type of attack, messages can be transmitted by a malicious node that has imitated another authentic one, by mimicking the latter's identity. In this way, the imitating attacker can send out malicious messages to any node by changing its own identity. The IoT node that has disguised its identity in this attack can capture authentic and secured packets, which can be disastrous for the entire network.

An application level attack: This type of attack in the IoT environment tampers with the messages and retransmits them to an untrusted non-secure destination. For example, in Internet of Vehicles (IoV) based implementations, the high-traffic lane could get maliciously broadcast as a 'congestion-free lane', leading to chaos during rush hour.

The importance of IPv6 in IoT

Security and integrity are the main issues in an IoT based network environment where interception-free secured communication is required. To enforce and integrate higher degrees of security, there is a need to implement IPv6 for IoT scenarios, with dynamic hybrid cryptography in key-generation and authentication. The IPv6 based approach can be enabled with fully secured algorithms, which are not vulnerable to interception. With the increasing implementation of IoT in diverse domains, it becomes necessary to work out the security aspects of IoT, by ensuring the secured routing of packets to prevent intrusions and ensure all transmissions can be fully secured.

RPL (Routing Protocol for Low-power and Lossy networks)

RPL is the IPv6 based protocol for IoT. It is primarily integrated for IPv6 over low-power wireless personal area networks (6LoWPAN). It works with the dynamic creation of Destination-Oriented Directed Acyclic Graph (DODAG), and has unidirectional as well as bidirectional communication. It has multiple instances with localised behaviour for higher optimisation. RPL enables each node in the framework to choose whether packets are to be sent upwards to their root or downwards to their child nodes.

Programming RPL with IPv6 in Cooja

Contiki is one of the widely used IoT enabled operating systems under free and open source distribution and is available at <http://www.contiki-os.org>. Contiki is equipped with the Cooja simulator, which is used for simulation as well as the programming of sensor devices. It offers enormous options to program the IoT nodes for real life implementations. The Cooja simulator enables the

programmer to import and program enormous types of IoT motes, and to get the results from different algorithms. To program, control and monitor the remote IoT devices, the backend C programs and related header files can be customised and recompiled to get the desired results. Contiki works on IPv4 as well as IPv6 networking with the integration of lightweight protocols, so that low power chips and radio frequency chips can be connected without performance issues.

Figure 1 depicts the simulation of multiple IoT motes in the Cooja simulator with the integration of dynamic cryptography. The simulation results can be analysed from the *Mote Output* in which there is an option to fetch the execution logs. All the data communication and signals are logged in *Mote Output* which can be further investigated using different types of graphs.

Once the simulation is complete, the network log files are analysed. This includes the source and destination motes, the time and the overall activities performed during simulation. In the *Mote Output* window, the log data can be copied and further analysed using data mining and machine learning tools for predictive analytics.

Figure 2 represents the *Collect View* in the Cooja simulator. It analyses the individual IoT mote. The IoT mote analytics can be viewed from the *Tools* menu of Cooja. From *Collect View*, any of the IoT motes can be analysed on multiple parameters including latency, temperature, time and other important parameters, which contributes to better performance evaluation.

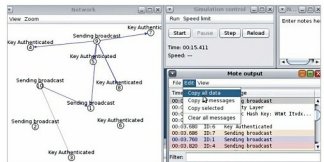


Figure 1: Fetching the data and messages for plotting graphs

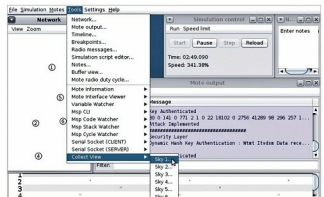


Figure 2: *Collect View* activation for detailed logs